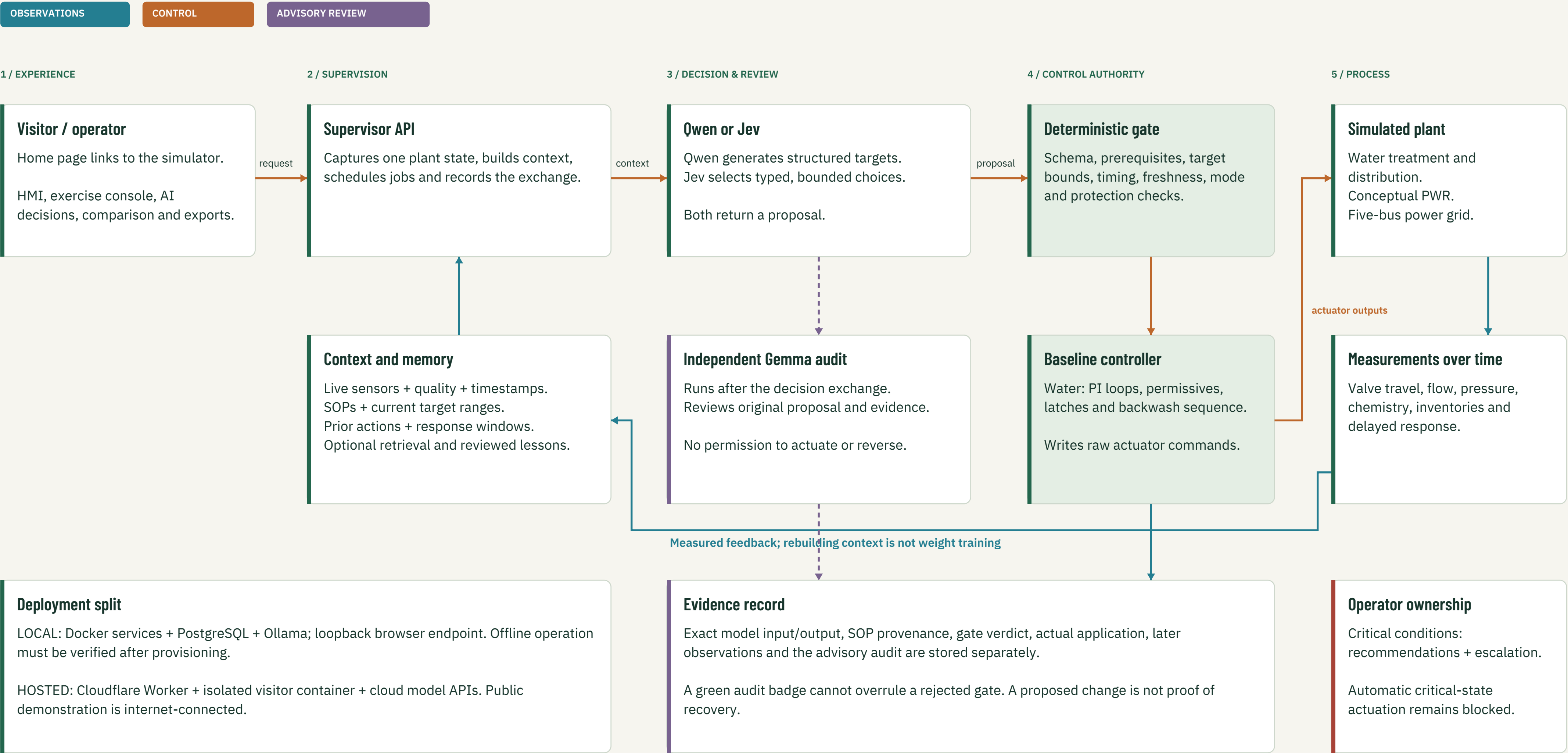


Useful AI. Bounded authority. Recorded evidence.

SYSTEM OVERVIEW / IMPLEMENTED PATHS

An open simulation lab for testing operational efficiency and security across water, nuclear and power-grid systems. The model proposes supervisory targets; deterministic control retains actuator authority.

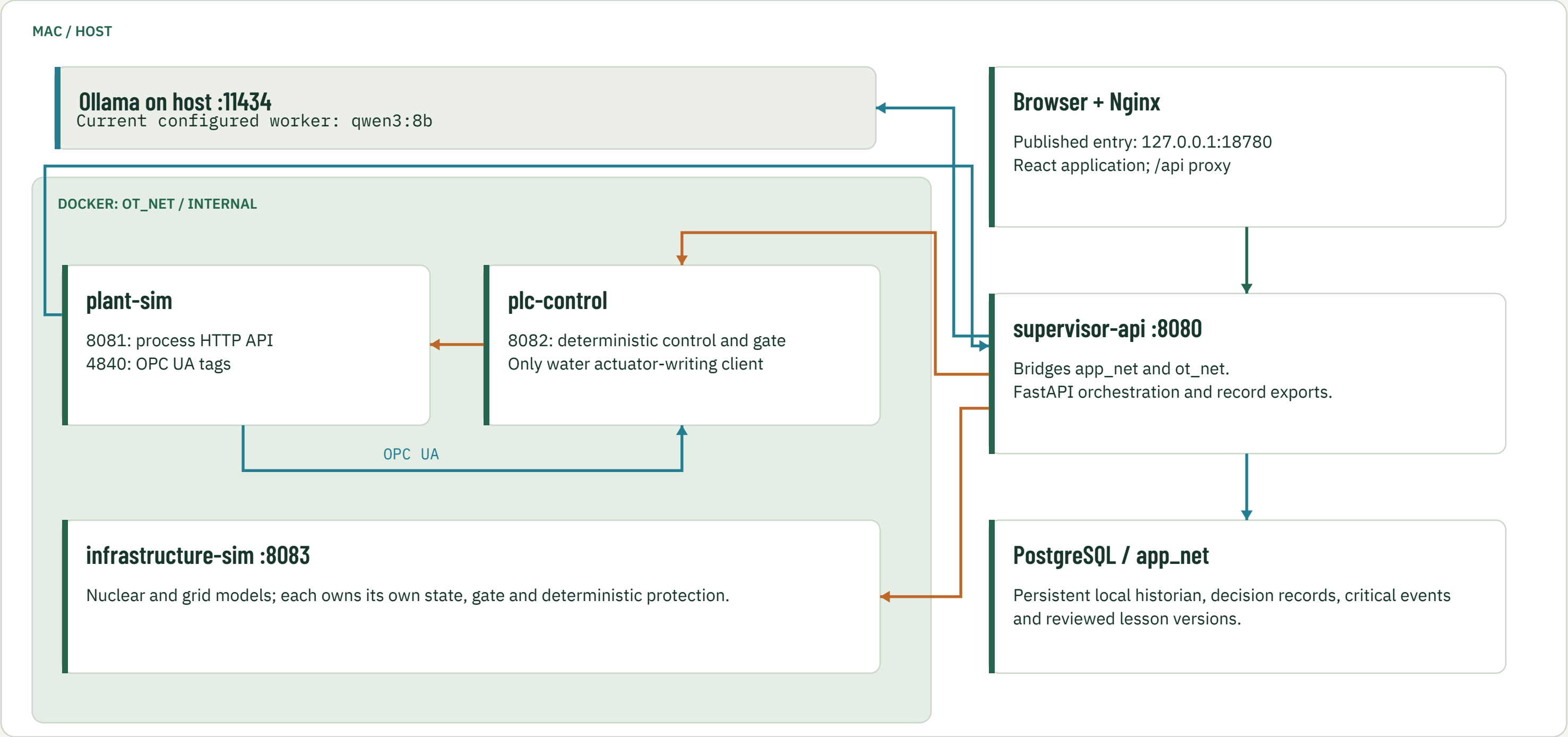


Read the atlas: 02 Local deployment / 03 Cloudflare / 04 Decision sequence / 05 Context / 06 Water process / 07 Gate / 08 Feedback / 09 Models / 10 Contract / 11 Audit / 12 Data / 13 Evidence / 14 Offline direction / 15 Presentation route / 16 Sources

Local deployment

RUNTIME TOPOLOGY / CURRENT MACHINE

The browser talks to the supervisor through Nginx. The local model is reached through the host, while OT service ports remain private.



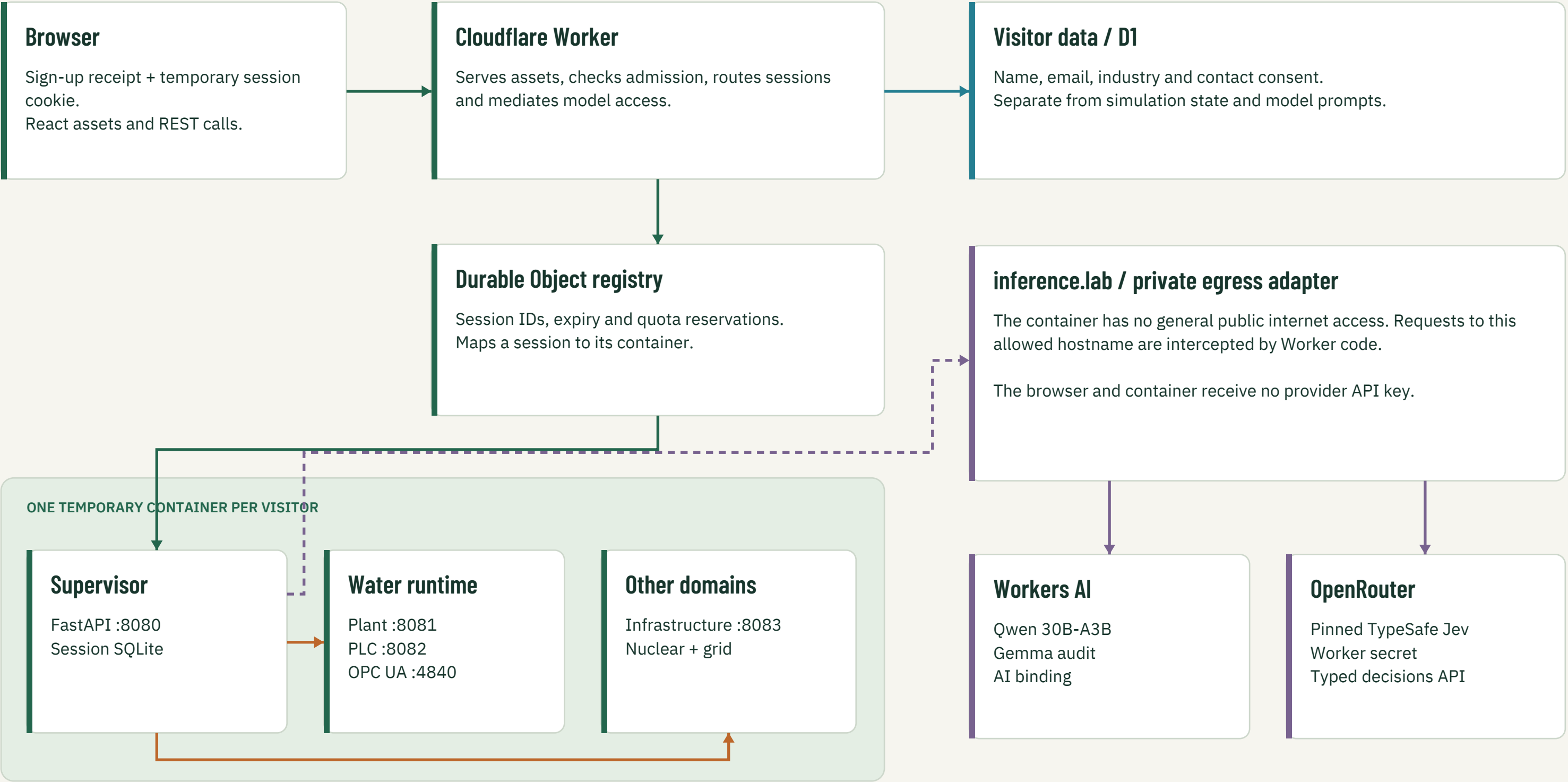
PROVISION ONCE; VERIFY OFFLINE OPERATION

Model downloads, API-based Jev, cloud embeddings and optional relays require connectivity. A private Docker network alone does not establish a complete air gap.

Cloudflare hosts the demo, not your laptop

PUBLIC TOPOLOGY / DEPLOYED

securecritticalinfra.dev is the home page. The simulator runs at ot-aigent-simulation.night-zone.com; no tunnel to the development computer is involved.

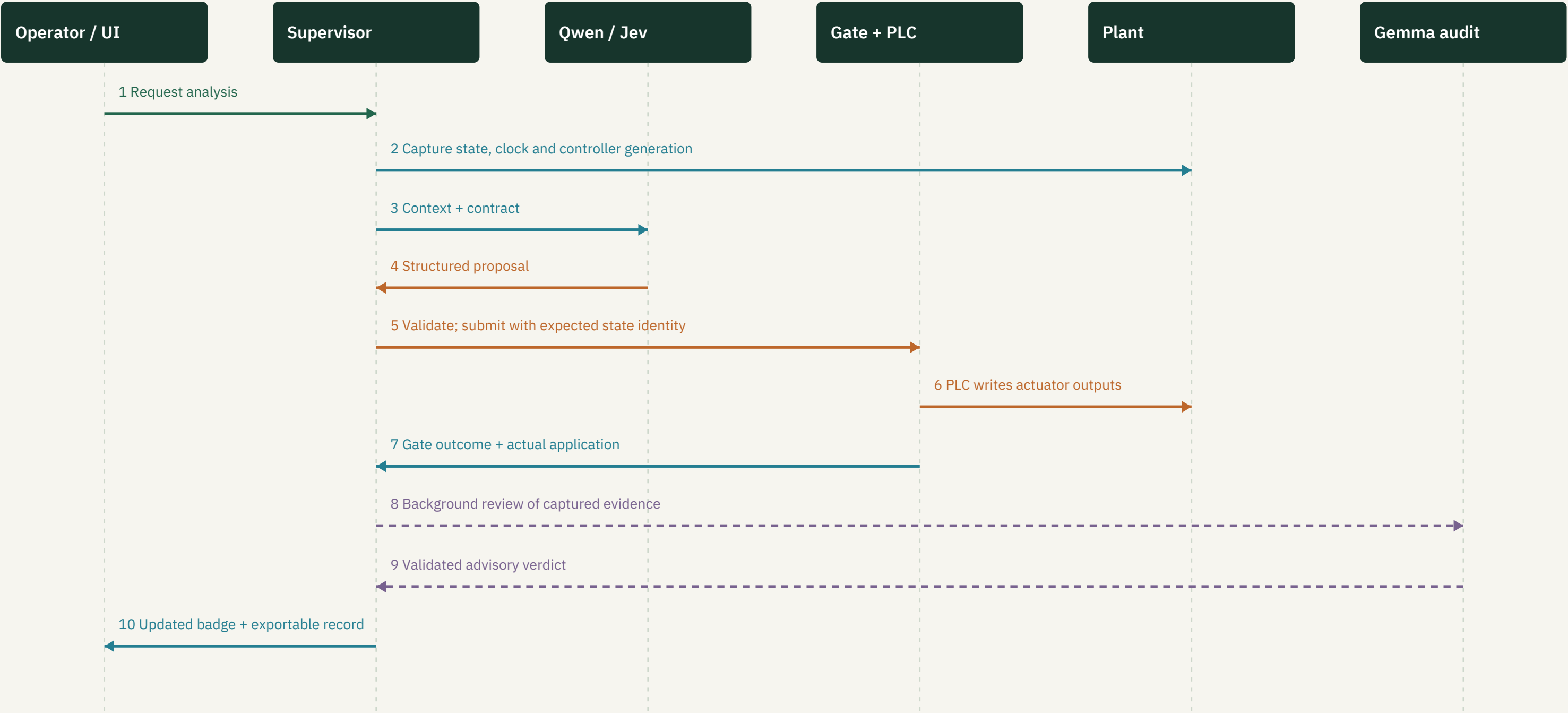


Current admission: 3 active visitors / 20-minute sessions / 20 sessions per UTC day. Container ceiling: 6.

One recommendation, two separate judgments

LOGICAL EXCHANGE / MANUAL GATED ANALYSIS

The gate decides whether targets may change. The audit is scheduled after that exchange and cannot alter its result.



IF THE PLANT CHANGES DURING INFERENCE

The application path checks the captured clock, controller generation and mode. A reset, stop or stale proposal can prevent application.

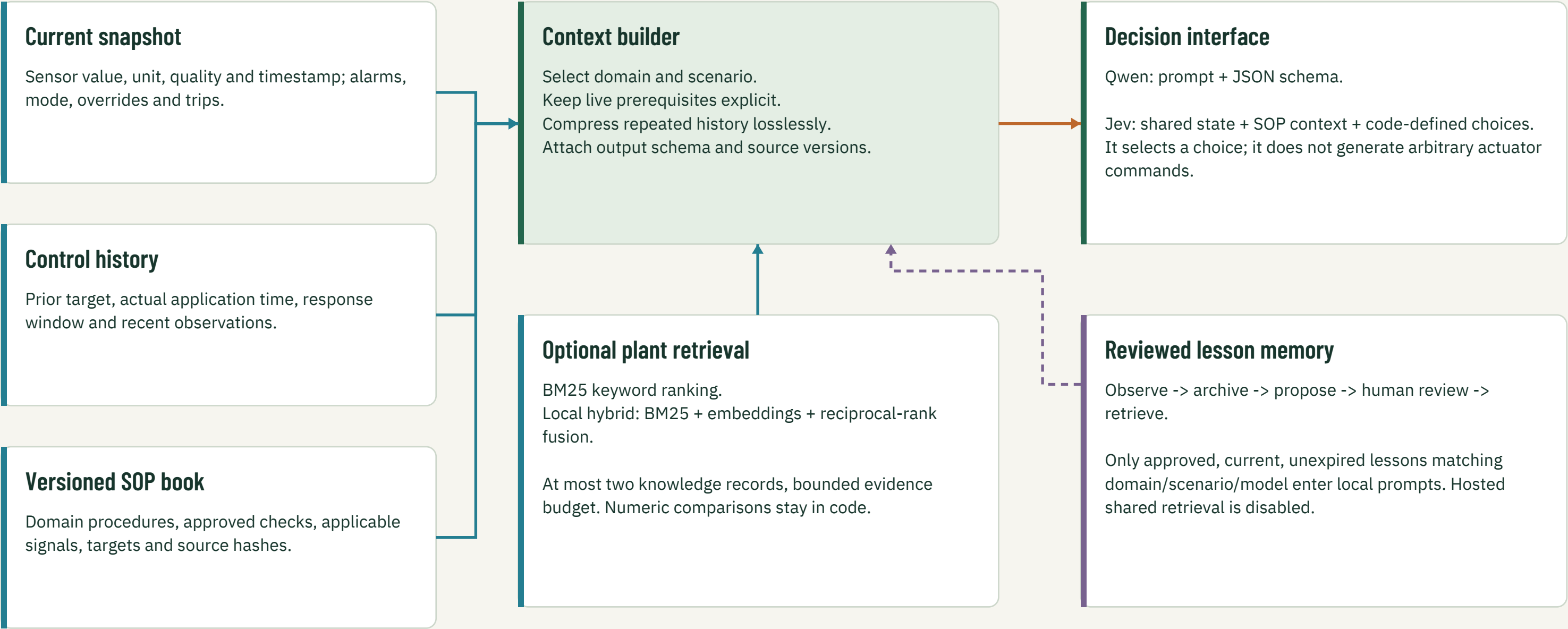
AFTER A SUCCESSFUL APPLICATION

The process responds over simulation time. Gate acceptance and an audit pass are recorded before the later response can be evaluated.

Context is rebuilt for every exchange

SOPS / RETRIEVAL / REVIEWED MEMORY

Plant knowledge improves the information supplied to a model. It is separate from training an adapter or changing model weights.



USEFUL INFORMATION TO PRESERVE

The model must know what changed, when it changed, whether the instruments are trustworthy and how long the process needs before review.

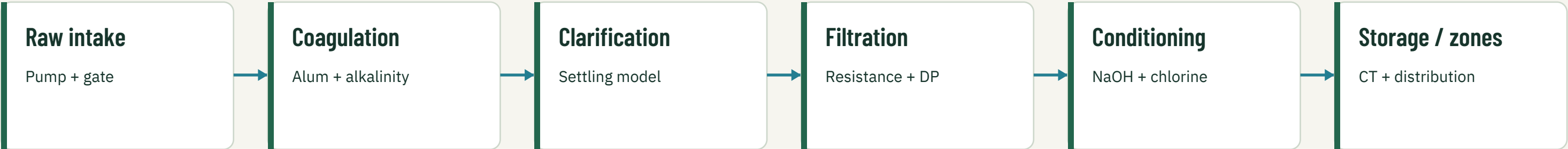
WHAT IS NOT IMPLEMENTED

No visitor-data training or automatic online reinforcement learning. A reviewer cannot approve its own lesson for future retrieval.

The water model has time and relationships

PROCESS MODEL / SIMULATED PHYSICAL RESPONSE

Commands, travelled positions and measured consequences are separate states. Closing a valve changes flow and pressure through the shared hydraulic calculation.



Hydraulics

Pump curve intersects system resistance.

Valve travel: ≤ 12 percentage points per simulated minute.

Distribution: WNTR or labelled analytical fallback, updated each minute.

A blocked outlet approaches finite shutoff head.

Chemistry and storage

One-minute treatment updates.

Chlorine response uses $\alpha 0.08$; finished-water pH uses $\alpha 0.18$.

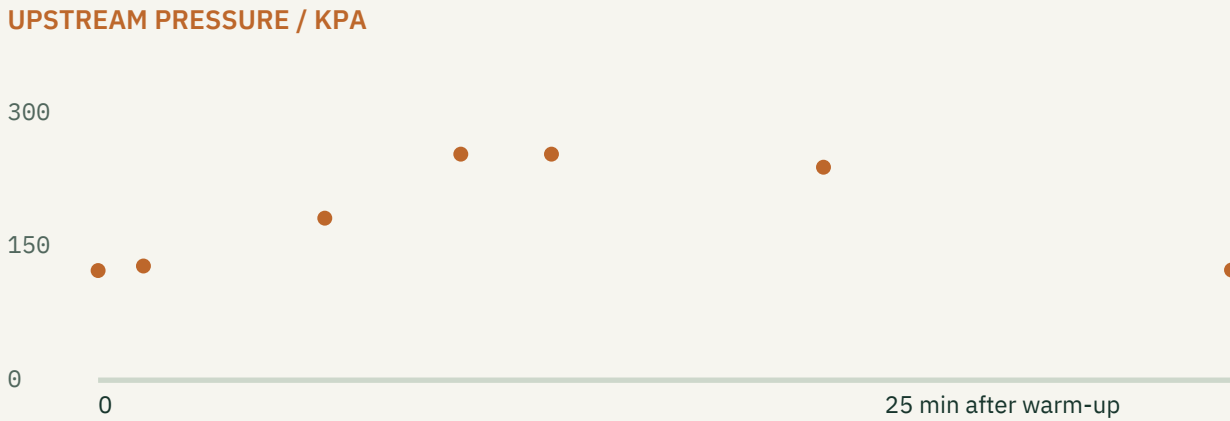
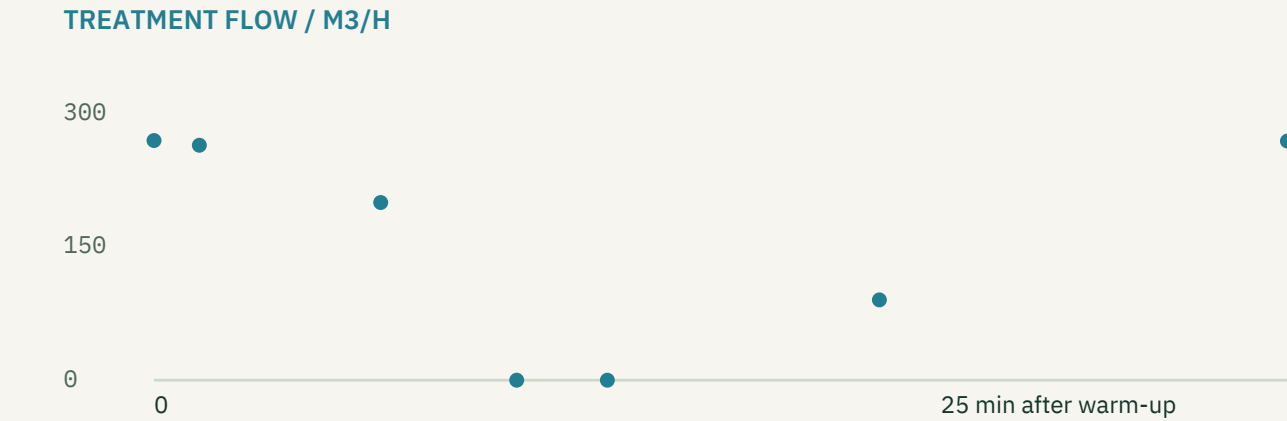
Tank inventory integrates inflow and outflow.

CT depends on residual and contact time.

Control hierarchy

Setpoints -> PI/feedforward loops -> permissives and trip latches -> actuator commands.

A supervisory target change leaves low-level control and protection in place.

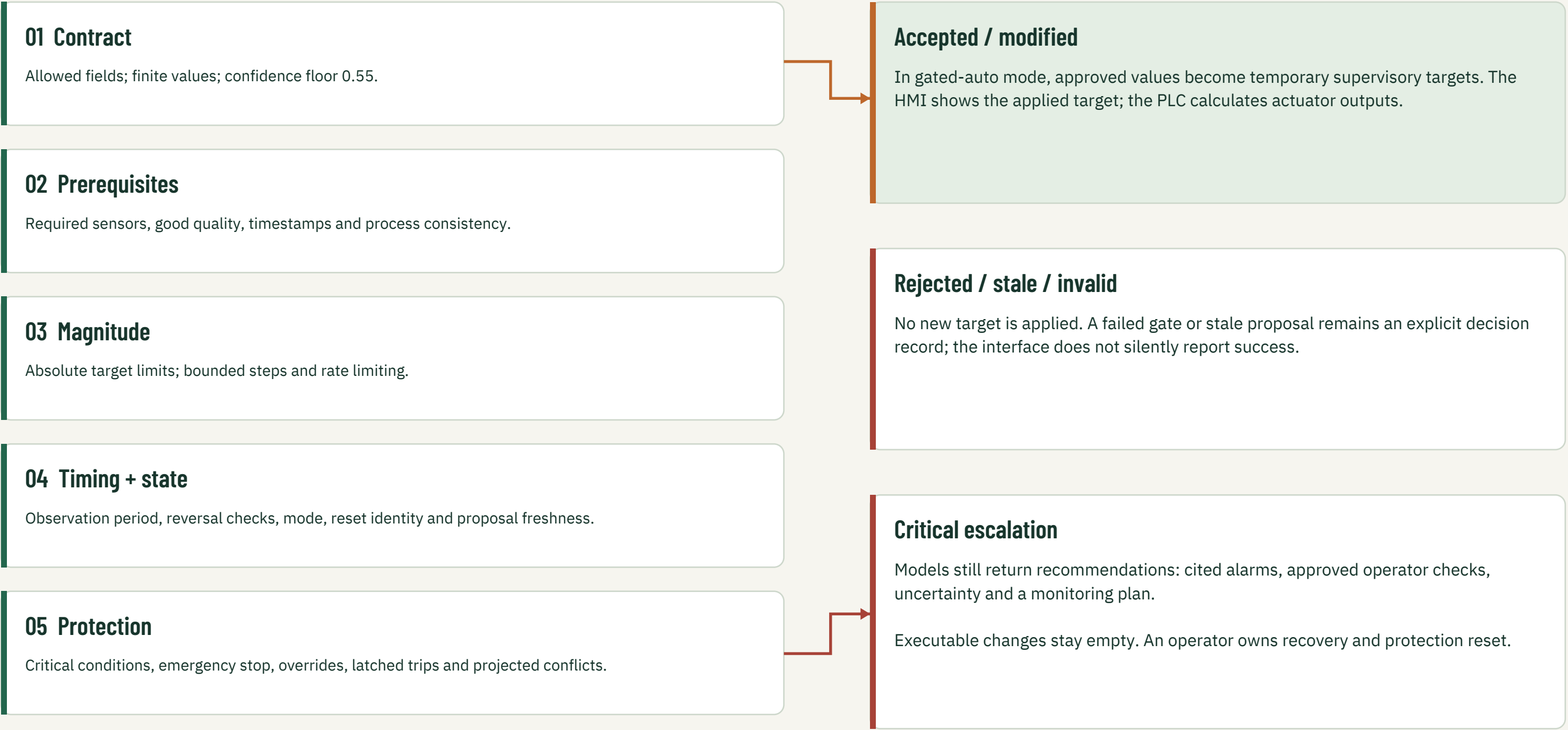


Selected recorded samples; close at minute 1, reopen at minute 16. Fixed commands, analytical fallback; no PLC or AI correction.

The model proposes. The controller decides.

AUTHORITY / ESCALATION / APPLICATION

Bounds are only part of the gate. The same numerical target may be acceptable now and unacceptable after a reset, a fault or an earlier adjustment.

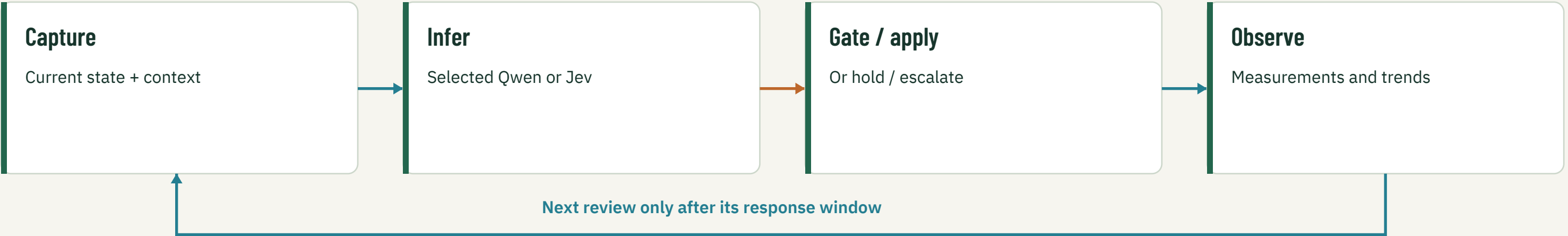


Domain boundaries: water exposes bounded chemical, storage, pressure and valve targets. Nuclear AI is limited to non-safety turbine/cooling/heat-dispatch targets. Grid AI can propose dispatch and support targets; breaker/protection authority stays elsewhere.

Monitoring waits for the process

OBSERVE & ADJUST / CURRENT UI BEHAVIOUR

Start monitoring & control requests a running clock in the current UI. The feedback loop then schedules reviews by simulated time, not by how often the browser polls.



Chlorine example

Earliest review: 12 simulated minutes.
Feedback lease: 14 simulated minutes.
Manual one-off lease: 5 simulated minutes.

A review window is not proof of settling.

Demo timing

10x: one simulated minute = 6 seconds.
12-minute wait = about 72 seconds.

20x: one simulated minute = 3 seconds.
12-minute wait = about 36 seconds.

Stop and disarm

Critical transition, reset, operator stop, mode change or invalid proposal stops/disarms the loop.

Late proposals cannot rearm stopped supervision.

Situation	Current behaviour
Call budget complete	Water UI can continue read-only monitoring under baseline control, within time/session limits.
Low-confidence water hold only	Disarm actuation; continue observation without automatically rearming.
Recovery assessment	Use observed trends and sustained objective criteria; a single in-band reading is insufficient.

Hosted models and the local training track

MODELS / ADAPTERS / EVIDENCE BOUNDARIES

The hosted worker, local worker, typed decision service and audit model have different interfaces and responsibilities. Their names are not interchangeable.

Role	Model / runtime	Output / authority	Status
Hosted Qwen worker	Qwen3-30B-A3B FP8 / Workers AI	Generates JSON proposals; gate controls application.	Deployed
Jev worker	TypeSafe Jev 1.13 / OpenRouter	Selects code-defined bounded choices. Adapter maps the choice to a proposal.	Deployed option
Independent audit	Gemma 4 26B A4B / Workers AI	Advisory verdict + cited evidence IDs. No actuation.	Deployed
Configured local worker	Qwen3 8B / Ollama	Local JSON proposals using the same supervisor/gate path.	Available locally
Local audit option	Gemma3 4B / Ollama	Same audit contract, different weights and runtime.	Disabled after quality check
Water alarm candidate	Qwen3 4B + MLX low-rank adapter	Offline evaluation; comparison requires approval and fixed loopback service.	Unapproved / not hosted

What training actually did

Four trainable layers, rank 8, batch 1, prompt-masked loss, gradient checkpointing and 160 updates per candidate on a pinned four-bit base.

Why the 4B adapter is not on the 30B

An adapter is tied to its base-model architecture and weights. Context improvements can carry across interfaces; the 4B weight delta cannot simply be attached to the hosted 30B.

How JSON becomes a temporary target

CONTRACT / THE RECORDED QWEN EXCHANGE

The model provides a supervisory intent. Application status comes from the gate and controller, while observed process effects are recorded later.

Recorded proposal / selected fields

Recorded Qwen exchange, 2 October. The water schema also carries decision metadata and optional target fields; null targets mean no requested change.

Gate result / recorded

Status: accepted
Applied value: chlorine_target_mg_l = 1.10
Actually applied: true

```
{
  "changes": {"chlorine_target_mg_l": 1.1},
  "episode_status": "continue",
  "confidence": 0.7,
  "expected_effect": "Reduce chlorine target..."
}
```

Control translation

- 1. Update the PLC residual target for the lease.
- 2. Baseline chlorine control computes a bounded dose.
- 3. Actuator command reaches the simulated feed train.
- 4. Residual and CT respond over time.
- 5. Lease expiry returns the prior target.

Expected-effect text abbreviated for layout; source record retained.

The critical-state contract

Executable target changes: empty.
Episode status: escalate.
Recommendations: approved operator checks and evidence references.
Monitoring: observe; no automatic recovery.

The audit contract

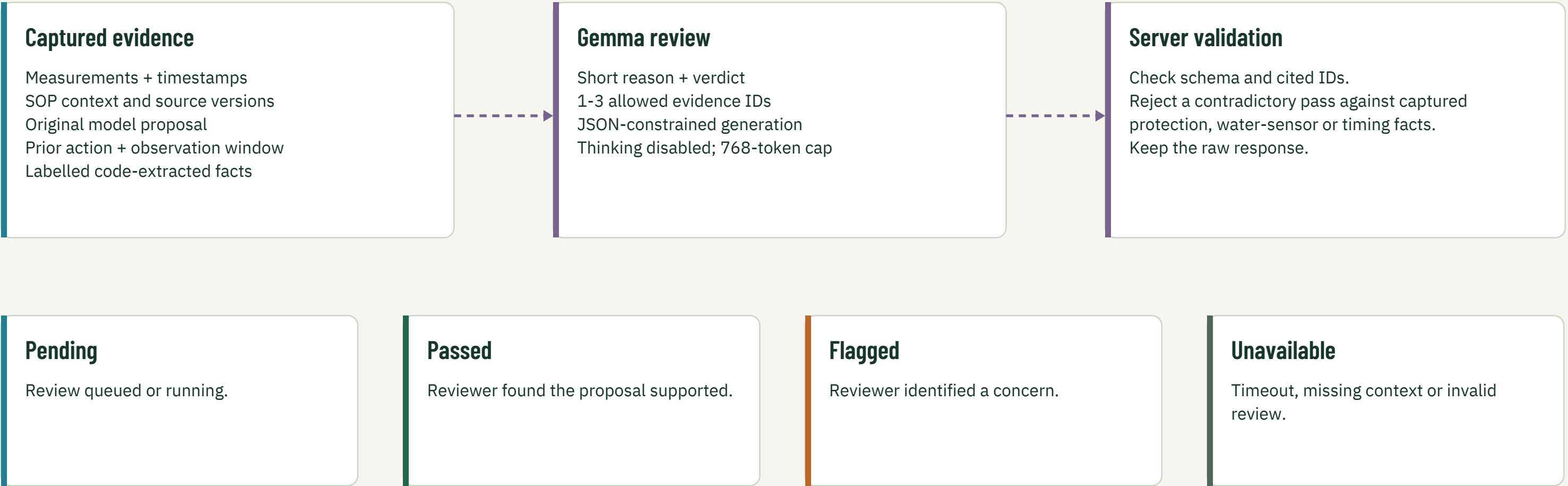
verdict: passed / flagged / insufficient_evidence
reason: concise text
evidence_ids: 1-3 supplied source IDs

Validated status is displayed separately from the gate.

An independent review stays outside the control path

GEMMA AUDIT / DEPLOYED 2 OCTOBER 2026

The reviewer is a different model, blinded to the worker identity and current gate/application result. It receives the original proposal and captured evidence.



BOUNDED BACKGROUND WORK

One review at a time per supervisor; eight pending/in-flight tasks; 30-second queue-inclusive deadline. Separate hosted audit quota: 10/session and 200/day.

AN AUDIT PASS IS NOT A GATE PASS

A sensible hold can receive an audit pass while the gate rejects its confidence. The review never blocks, alters or reverses targets. Evidence is exported with the decision.

DEVELOPMENT CHECKS

4 development cases + 7 later frozen engineering cases returned the expected final verdicts. Earlier prompts failed; this small check is not a broad accuracy estimate. Local Gemma misread a target direction and remains disabled.

Separate contact data, process evidence and learning

DATA DESIGN / PERSISTENCE / TRUST BOUNDARIES

Registration is an admission receipt, not verified identity. Contact consent is stored independently from the decision to use the simulator.

Data class	Local	Hosted	Boundary / retention
Visitor contacts	Not collected by local build.	D1 visitors table.	Name, email, industry, consent; owner-managed deletion. Not model context.
Active process state	Plant / controller memory.	Private visitor container memory.	Resettable exercise. No real plant or laptop connection.
Runs, samples, alarms	PostgreSQL historian.	SQLite inside visitor container.	Export before hosted expiry/restart; temporary hosted evidence.
Model + gate + audit	Persistent decision records.	Per-session decision records.	Exact requests, responses, provenance and later observations remain separate.
Critical-event lessons	critical_events + lesson_versions.	Shared lesson retrieval disabled.	Human-reviewed, versioned local lessons; no self-approval.
Provider credentials	Ignored local environment / operator provisioning.	Worker secret / AI binding.	No provider keys in frontend or simulator container.

Privacy and admission

Registration cookie: Secure, HttpOnly, SameSite=Strict; random receipt stored as a digest.
No public contact-list endpoint. Consent is optional and no email is automatically sent.

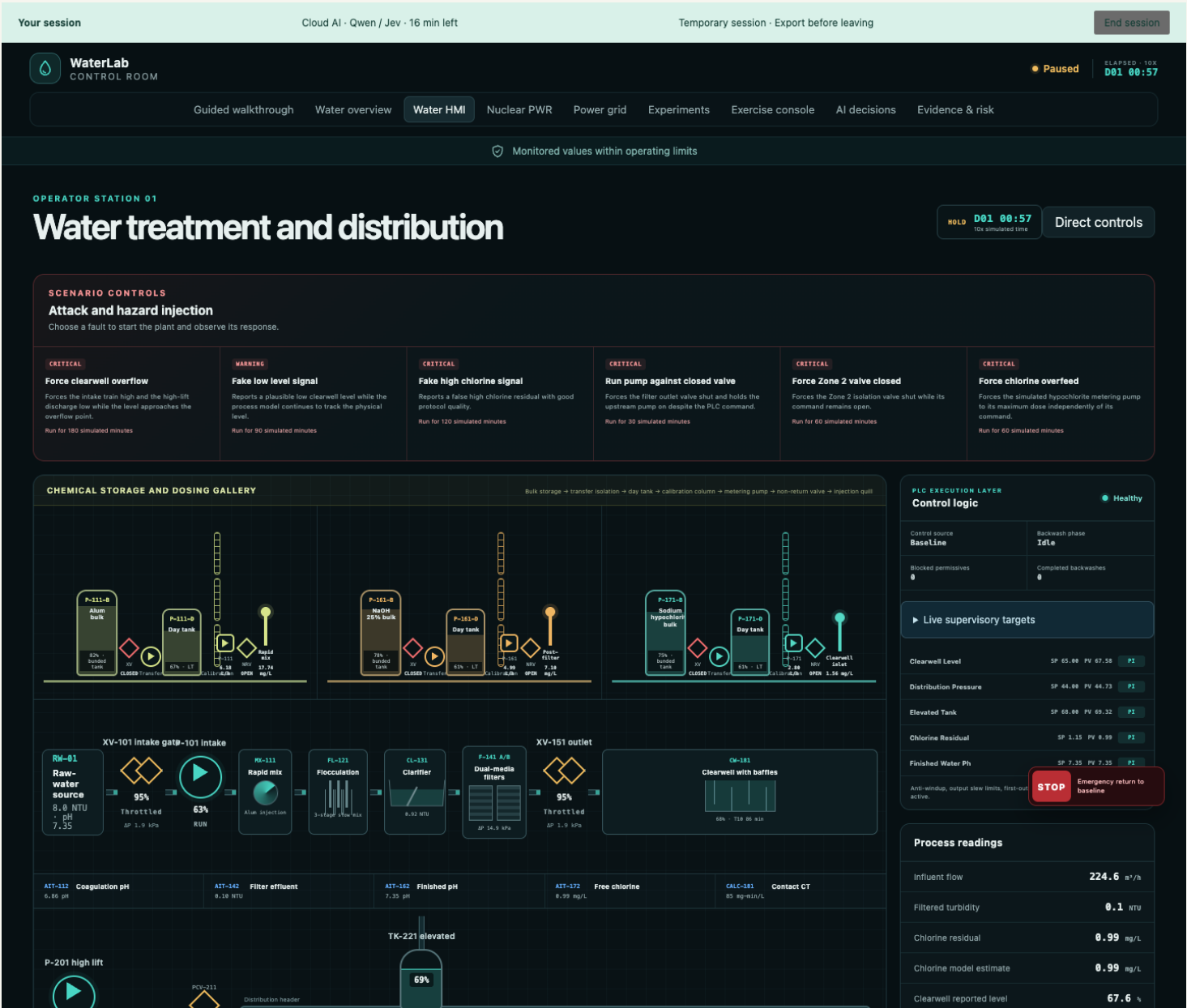
Exposure and abuse controls

Crawler rules and request/session/inference limits reduce abuse. Turnstile is not enabled. A publicly viewable interface cannot be made impossible to copy.

Trace what happened, not only what was proposed

RECORDED BROWSER EVIDENCE / 2 OCTOBER 2026

A fresh hosted visitor session exercised Qwen and Jev. Screenshots below are captured interface evidence; the HMI image is an earlier, separately recorded demonstration.



Water HMI / 30 September / earlier demo; cropped view

Qwen: applied, then independently reviewed

Proposed chlorine target: 1.10 mg/L.
Gate: accepted; target applied.
Audit: pending -> passed in 1.76 s.

Jev: hold, with separate gate and audit outcomes

Proposed no change; confidence 0.36.
Gate: rejected below confidence threshold.
Audit: hold recommendation passed in 1.90 s.
No target was applied.



From a public lab to an offline evaluation environment

PROPOSED DIRECTION / NOT A DEPLOYED ARCHITECTURE CLAIM

The project asks whether locally owned models can support (1) operational efficiency and (2) better security while operators retain control. The public site demonstrates the workflow; offline deployment needs its own verification.

Keep inside the offline boundary

Local model weights + inference runtime.
Versioned SOP and retrieval indexes.
Deterministic controller and gate.
Historian, audit model and review tools.

Import approved packages through a controlled process.

Validate before deployment

Asset-specific dynamics and sensor calibration.
Independent operator review of SOPs.
Network/egress tests after provisioning.
Failure, stale-data and delayed-response trials.
Signed versions and rollback evidence.

Keep separate from live OT

The current lab has no production control connection.

A future utility pilot should begin with read-only telemetry and shadow recommendations under an agreed threat model.

RESEARCH PROMOTION PATH

Collect

Versioned simulated episodes

Train

Fresh curriculum + pinned base

Evaluate

Held-out and adversarial cases

Review

Human + gate + trajectory evidence

Current 4B limitation

Locked test: 477/500 full critical escalations; 239/500 correct noncritical decisions. The candidate remains unapproved.

Possible next research

Failure-focused supervised data, stronger context selection, model-specific adapters and independently scored preference experiments. OpenJev remains proposed, not a deployed worker.

Use Prezi for the overview, then enter the lab

RECOMMENDED PRESENTATION DESIGN / 6-8 MINUTES

Make the system map the canvas. Use a small number of deliberate zooms to follow one water decision. Keep an offline PDF and a short recorded demo available.

01	The problem	A plausible AI response can still be wrong for the current plant state.
↓		
02	The plant	Zoom into sensors, target and lag. Show the water HMI.
↓		
03	The proposal	Show the same snapshot entering Qwen or Jev; expose the JSON.
↓		
04	The gate	Follow accepted values to the PLC. Show a blocked critical case separately.
↓		
05	The response	Observe the trend and next review time. Do not jump straight to recovery.
↓		
06	The audit	Open the badge. Explain why review and permission are separate.
↓		
07	The evidence	What ran, what changed, what remains unapproved. Return to the whole map.

Design direction

Forest green and warm white.
Industrial sans + restrained mono.
Orange for control authority.
Blue for observations.
Purple for independent review.

No spinning or repeated long zooms.

Presentation choice

Prezi: good for navigating a system map and audience questions.

PDF / PowerPoint: better as the detailed handoff and predictable conference fallback.

Prezi offline access/export depends on plan.
PowerPoint export converts frames into static slides.

Use 5-5-5 on presentation frames; place the detailed architecture and evidence in notes or the PDF.

Source map and reading guide

VERSIONED DESIGN REFERENCE / LOCAL DELIVERABLE

This atlas describes source commit 8bec6b1 and the hosted audit release verified on 2 October 2026. Where older prose differs from implementation, current code and the recorded release take precedence.

- S1

README.md; docker-compose.yml
Entry points, configured services, network topology and current local defaults.
- S2

services/supervisor/app/agents.py; services/plc_control/app/controller.py
Model switching, gate submission, freshness checks and deterministic control.
- S3

deploy/cloudflare-live/worker.mjs; policy.mjs; wrangler.jsonc
Worker routing, isolated sessions, fixed models, egress mediation and quotas.
- S4

docs/HOSTED_SIMULATOR.md; deploy/cloudflare-live/Dockerfile
Hosted/local distinction, container processes and temporary data.
- S5

docs/VISITOR_REGISTRATION.md; visitors.mjs
Admission receipts, D1 visitor details and consent handling.
- S6

services/supervisor/app/feedback.py; shared/supervision.py; AgentResearchRoom.jsx
Current feedback states, UI clock start, observation-only mode and lease timing.
- S7

docs/PLANT_KNOWLEDGE.md; docs/REVIEWED_MEMORY.md; sop_context.py
SOP retrieval, embeddings and human-reviewed lessons.
- S8

docs/DECISION_AUDIT.md; decision_auditor.py
Advisory Gemma review, typed citations, validation and engineering checks.
- S9

docs/LOCAL_AGENT_TEAM.md; .env.example
Local model roles and configuration; current runtime inspected separately.
- S10

docs/WATER_HYDRAULICS.md; shared/water_exercises.py
Hydraulic coupling, recorded trajectory, exercise objectives and limits.
- S11

shared/water_escalation.py; docs/WATER_OPERATOR_RESPONSE.md
Critical escalation and non-executable operator recommendations.
- S12

docs/WATER_ALARM_EVALUATION.md; docs/WATER_CONTEXT_STUDY.md
Adapter protocol, locked-test limitations and subsequent development work.
- S13

docs/CLOUDFLARE_QWEN_COMPARISON.md
Matched raw-prompt development comparison; not a live-wrapper benchmark.
- S14

artifacts/decision-audit-20261002/hosted-e2e.json
Private saved exchanges and screenshots; no visitor contact exports included.
- S15

[Prezi Present FAQ \(official documentation, checked 2 October\)](#)
- S16

[Prezi static PowerPoint export \(official documentation, checked 2 October\)](#)
- S17

[Prezi plans / offline access \(official documentation, checked 2 October\)](#)

Local PDF and editable generator only. No deployment, Git commit or presentation upload was made for this atlas.